Micro Alphas

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Abstract

We introduce the concept of micro alphas, weak signals that may not reach statistical significance individually, but when combined, generate predictability for the equity risk premium. Unlike previous predictability studies born out of the Goyal and Welch (2008) critique, predictors do not have to exhibit statistical significance consistently over the sample for them to be considered. Though the recent update from Goyal, Welch, and Zafirov (2024) is more nuanced than the original, it still evaluates candidate variables against an only partially relevant criteria. Borrowing from the machine learning literature, we further apply transformations and feature selection in a cross-validated walk-forward way, in order to capture potentially non-linear and time-dependent effects. Using an elastic net model, we present a strategy demonstrating consistent excess returns over the S&P 500, and discuss methods for continually improving the model.

Keywords: Equity risk premium; market timing; tactical asset allocation; predictability; weak signals; machine learning

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1 Introduction

Classical finance theory is grounded in the idea of the efficient market hypothesis (Samuelson 1965; Fama 1970). While it is tautological that investors cannot *all* beat the aggregate market portfolio, both academics and practitioners have had a long, open debate about whether *some* individual investors might be able to outperform on a sustained basis that would be consistent with skill rather than luck. According to the seminal Goyal and Welch (2008) paper and the so-called "Goyal-Welch critique", the common predictors found in 30 years of previous literature fail to beat an historical average when taken out-of-sample. Most of the literature on investment funds shows significant evidence that most active managers underperform (at least once fees are taken into account), and that the overperformers cannot sustain their performance, and mean-revert over time (see Cremers, Fulkerson, and Riley 2019, for a recent survey). Yet large swaths of investors, including among the most financially literate, keep attempting just that, with apparent success (Hull and Qiao 2017). On the academic side, recent attempts have been made to rehabilitate predictability of the equity risk premium (ERP), notably using machine learning techniques (e.g. Kelly, Malamud, and Zhou 2024).

The recent Goyal, Welch, and Zafirov (2024) update to the Goyal and Welch (2008) paper provides an interesting look into how the academic view on ERP prediction has evolved since the original exercise. Their critique of data mining and in-sample results is now a mainstream part of empirical research, with new predictors now generally having to provide a theoretical rationale for why they are predictive. Given the potential for data snooping, the discussion on significant t-statistics is now prominent, along the lines of Harvey, Liu, and Zhu (2016) for the cross-section of returns. The updated results of Goyal, Welch, and Zafirov (2024) are actually a lot more nuanced than in the original paper, finding a few predictors that in fact appear fairly robust in their extended sample, but their conclusion remains skeptical. Importantly, their methodology, and that of many of the top publications they consider, still relies on the assumption that they are looking for strong predic-

tors likely to survive after publication, despite persuasive reasons to expect the opposite (McLean and Pontiff 2016).

In this paper, we make a practitioner's case that there are many weak, noisy signals that correspond to "micro alphas" that can provide significant overperformance over time. Our methodology identifies and aggregates these individually weak predictive signals and combines them to create a substantial predictive edge. Drawing from machine learning principles, we use an elastic net regularization approach that effectively balances the selection of relevant predictors while managing multicollinearity and overfitting issues inherent in financial data. By fitting this model monthly within an expanding-window framework, we avoid potential look-ahead bias and enhance the robustness of our predictive signals. A critical component is the use of feature transformations, which are optimized in-sample, including key indicators of sentiment, volatility, and economic conditions, to capture shifting patterns in the equity risk premium (ERP) and market dynamics over time.

Our approach is in line with the concept of weak learners, common in the machine learning literature, where models that perform only slightly better than chance, and can be combined in an ensemble approach to provide stronger predictability (Freund and Schapire 1997). In a finance context, the idea of strong alphas has long been antithetical to theoretical foundations such as market efficiency and no-arbitrage pricing. The micro alpha approach can then be justified along three broad, non-exclusive categories: limits to arbitrage; behavioral phenomena; and dynamic/adaptive market equilibrium, in the vein of Lo (2004).

We find that our methodology successfully identifies micro alphas, leading to an out-of-sample strategy that outperforms the market despite similar risk characteristics. Our empirical results consistently outperform the market benchmark over a range of market conditions. The strategy's overall out-of-sample R^2 is approximately 1.33%, indicating a non-trivial predictive relationship with daily returns. This predictive strength is particularly pronounced during periods of heightened market stress, such as the 2008 financial crisis and the COVID-19 pandemic, where the model's out-of-sample

R^2 peaks. Throughout most of the sample, the rolling R^2 remains above zero, suggesting that the strategy persistently generates signals that add predictive value across diverse economic environments. Notably, periods of high model performance align with medium to high volatility in the VIX index, reinforcing the idea that market turbulence enhances the effectiveness of our predictive signals. This aligns with our theoretical priors: in times of higher uncertainty, inefficiencies are more likely to occur and for us to capture. Conversely, it is hard to find alphas that are resilient in periods of relative calm.

While the literature often focuses on prediction errors and out-of-sample R^2 as a measure of skill, it is worth stressing that they bear relatively little relation to the earnings that would incentivize a trader to implement such a strategy, and therefore contribute to market efficiency. Therefore, it is critical to evaluate whether a simulated portfolio beats the benchmark on a risk adjusted basis. Although this result is to some extent dependent on the choice of a portfolio allocation rule, returns are of far greater relevance than R^2. A statistic that measures the linear association of two normally distributed variables is not rich enough to completely describe any realistic market timing model.

Our strategy's performance metrics indeed confirm the effectiveness of aggregating micro alphas. With a full-sample Sharpe ratio of 0.556, Sortino ratio of 1.074, and an annualized alpha of 6.4% (closer to 2% after 2010), the strategy demonstrates substantial risk-adjusted returns compared to a traditional equity benchmark. Maximum drawdown for the strategy is notably lower than the benchmark (38.5% vs. 55.2%), and time under water is reduced, underscoring the model's resilience during downturns. Feature importance analysis shows that volatility signals play a consistently strong role, while other features, such as macroeconomic indicators, contribute meaningfully during certain economic cycles. This dynamic feature relevance further supports the model's capacity to adapt to shifting market conditions. Additionally, the time-varying coefficients of our core predictors illustrate how specific indicators gain or lose influence depending on market phases, underscoring the strategy's flexibility in evolving with financial conditions and

maintaining a competitive edge over time.

In subperiod analyses, we confirm that the strategy maintains strong performance and adaptability across various market environments, with consistent outperformance relative to the benchmark. For the 2010-2024 and 2019-2024 subsamples, the strategy achieves even higher compound annual growth rates (CAGR), risk-adjusted metrics, and reduced drawdowns compared to the full sample, suggesting that it effectively capitalizes on post-crisis trends and heightened market volatility. The Sharpe and Sortino ratios improve in these subperiods, highlighting the model's ability to manage risk more efficiently in recent years. Additionally, the positive CAPM alpha (though reduced to around 2%) and increased hit rate during these periods affirm the strategy's capacity to generate excess returns that are greater than a passive exposure to the market beta. These findings suggest that the model's combination of weak predictors is economically significant and adapts well to evolving market conditions, providing a reliable signal that enhances both returns and resilience during times of economic uncertainty.

The paper is organized as follows. Section 2 discusses the relevant literature and sets up our hypotheses. We then present our data and empirical methodology in Section 3, and discuss our results in Section 4. We present some robustness checks in Section 5 and discuss their significance. Section 6 concludes.

2 Related literature

2.1 Equity risk premium prediction

Although the literature on market efficiency goes back six decades, Goyal and Welch (2008) provides the modern framework for how difficult predicting the equity premium may be. Examining a wide range of variables that have been proposed in prior literature as potential predictors of stock returns, including dividend yields, earnings ratios, and interest rates, they find that in out-of-sample tests, virtually none of these variables can consistently outperform a simple historical average. No variable was economically significant enough to overcome even modest transaction costs. This suggests that any apparent predictability in stock returns was likely due to data mining or look-ahead bias, rather than genuine forecasting ability, aligning closely with the concept of strong-form market efficiency. While these results are striking, they look at a limited number of predictors and do not tackle the question of timing with weak predictors.

Hull and Qiao (2017) find they can forecast market returns 6 months into the future and implement a market timing strategy that outperforms a buy-and-hold approach, achieving annual returns over 12% with a Sharpe ratio of 0.85 from 2001-2015. The study concludes that market timing based on return forecasts can be profitable. The outperformance was in part due to the correct prediction of two major market downturns in the sample period.

The importance of predicting severe market declines is also emphasized in the next two papers. Li and Tsiakas (2017) find that the equity premium is predictable out-of-sample when using a "kitchen sink" regression conditioned on economic fundamentals, combined with economic constraints (signs of coefficients) and statistical constraints (shrinkage). This predictive framework outperforms the historical mean benchmark and other competing models, with stronger performance during recessions. They find technical indicators add little beyond the predictive content of economic fundamentals.

On a related note, Tsiakas, Li, and Zhang (2020) examine more specifically how equity premium prediction depends on the state of the economy.

The authors find that certain economic fundamentals (like the dividend yield) predict better in expansions, while others (like the short rate) predict better in recessions. They propose a simple forecasting approach that combines one cyclical and one countercyclical predictor, showing it achieves diversification and insurance benefits, performing well across different economic states.

Hull, Qiao, and Bakosova (2019) develop a one-month market timing model using a diverse set of macroeconomic and technical variables. Using weighted least squares with stepwise variable selection, they construct a predictive model for one-month-ahead market excess returns and transform these forecasts into investable positions. Their strategy substantially outperforms the S&P 500 in terms of both returns and risk-adjusted performance, while experiencing much smaller drawdowns during market downturns. When combined with their earlier six-month model, the resulting strategy performs even better, demonstrating the benefits of combining market timing models operating at different frequencies.

Revisiting their earlier work, Goyal, Welch, and Zafirov (2024) provide a more nuanced picture as of 2022, using 29 new predictive variables from 26 papers published after their 2008 paper, in addition to re-examining the 17 original variables. They find that more than a third of the new variables examined have lost their predictive power even in-sample when extended to 2021, and of those that retain in-sample significance, about half show poor out-of-sample performance. However, a small number of variables, including some from their original 2008 set, still perform reasonably well both in-sample and out-of-sample. All but three variables [1] fail to outperform a simple all-equity investment strategy in hypothetical trading scenarios. This pattern seems consistent with adaptive market efficiency, but still leaves a door open for market timing.

[1] Namely *accrul*, aggregate accruals from Hirshleifer, Hou, and Teoh (2009), *gpce*, fourth-quarter growth in personal consumption expenditures, and *gip*, the growth rate in industrial productions, both from Møller and Rangvid (2015).

2.2 Limits to market efficiency

In order to justify why we might be able to predict the ERP, we need to accept that there are practical limitations to strong-form market efficiency. We focus on three main perspectives: limits to arbitrage; the behavioral critique; and the adaptive market framework.

First, there are a number of macro and microstructure reasons why investors may not be able to fully take advantage of arbitrage, even when they are perfectly rational. This can be true of even risk-free arbitrage when the profits are low and the required capital is high and carries opportunity costs (e.g. Du, Tepper, and Verdelhan 2018). Transaction costs, direct (fees) or indirect (bid-ask spreads, execution quality), short-selling costs, and other capital and regulatory constraints fall under this umbrella (Shleifer and Vishny 1997; Gromb and Vayanos 2010). Furthermore, when arbitrage is statistical in nature and not risk-free, the risk/reward profile of a particular anomaly may make it unattractive or impractical to carry. Weak and temporary signals, in this context, may therefore be disregarded by arbitrageurs.

Second, the behavioral critique in economics and finance has provided fodder for the idea that limits to rationality may create predictable and exploitable patterns in investors' behavior. Proving irrationality is hard, and many phenomena, such as herding and runs, can often be reconciled in a rational expectations framework. However, there is a growing body of evidence converging towards showing that at least some investors repeatedly act in suboptimal ways given a reasonable utility function (Daniel, Hirshleifer, and Subrahmanyam 1998; Hirshleifer 2001; Hirshleifer, Lim, and Teoh 2009).

Along the same line, limited investor attention has been linked to under and overreaction to news, which may create return predictability (see Lim and Fteoh 2010, for a survey). While inattention is plausible when it comes to individual firms, the overall market's ERP by nature attracts many eyeballs. However, certain predictors may be irrelevant for such extended periods of time that even sophisticated investor stop including them in their models, or lack good data to predict their effect. For instance, inflation expectations were of little concern to most U.S. equity investors for many years,

but became of utmost importance after inflation's post-pandemic resurgence.

This connects to the third line of arguments, along the line of Lo (2004)'s adaptive market hypothesis (AMH), developed in the 30th anniversary issue of *The Journal of Portfolio Management*, which may mediate the previous two. In this framework, market efficiency is viewed as the dynamic outcome of an ecosystem of actors with different objectives, preferences, and horizons, interacting with evolving financial products with various levels of supply, affected by exogenous shocks of various kinds. Such a system is too complex to model realistically in a way that is tractable in a general equilibrium sense, but is a plausible representation of real-world financial markets. In particular, two practical implications of AMH are relevant for our approach to market timing.

The first implication is that the equity risk premium is time-varying and path-dependent[2], but in an evolutionary way. In zoological cycles, certain species' populations expand until they outgrow their habitat's capacity, or more predators come in and curb the growth. By analogy, investors may "crowd a trade" and compete away an anomaly that was once profitable; or, a trade with seemingly low risk may become riskier as other players discover it. Along these lines, it is useful to frame tactical investment decisions in an heterogeneous investor context: different agents have different risk preferences and objectives, and especially different life-cycle considerations. If the marginal investor tilts in favor of a younger investor with high-risk, high-reward preferences, as opposed to a retiree in a decumulation phase, market dynamics may look dramatically different.

The second implication is that some arbitrageable alphas can be found, at least temporarily, consistent with the Grossman and Stiglitz (1980) paradox. There needs to be some expected (aggregate) benefit to attempting the complicated process needed for making markets more efficient, or nobody will. Because of the dynamic nature of the equilibrium, because of the cycles, bubbles, panics that can result from it, capturing these alphas may require significant (and costly) innovation. Further, as in the evolutionary

[2]Which Lo points out can be obtained in rational expectations models, too.

context, there is no guarantee that any given attempt will be successful or that a particular strategy will be adapted to a changing ecosystem.

A key distinction further lies between cross-sectional and aggregate market returns. While cross-sectional anomalies can be arbitraged by institutions taking diversified long and short positions across many stocks, the equity risk premium offers only a single, undiversified bet. This structural limitation restricts the number of participants who can engage in market timing without assuming significant concentrated risk. As a result, inefficiencies in broad market predictability may persist longer than those in cross-sectional returns, creating opportunities for capturing overlooked signals that remain inaccessible to most diversified strategies.

3 Empirical analysis

3.1 Data

Core model. Our main model is built from 14 features which are the result of a variable selection process from a wide set of candidate features. For clarity, we perform our primary analysis on this selection only. We make no claim that our model is in any way optimal, merely that it shows the possibility of successful market timing.

Our core features span January 2, 1990 to June 6, 2024, and can be classified into four categories:

1. **Broad market and macro:**
   (a) *BAA*: yield for BAA from Moody BAA Index;
   (b) *bfc*: Bloomberg Financial Conditions Index;
   (c) *delinquency*: Federal Delinquency Rates for Commercial Banks & Industrial Loans;
   (d) *eyby*: SP500 Equal Weighted Median Earnings Yield (from Ned Davis Research) normalized by an average bond yield (simple average of BAA, US3M, and US10M)
   (e) *termspread*: term spread (10-year minus 3-month Treasury rates);

2. **Sentiment**:
   (f) *n265*: Ned Davis Research's Daily Trading Sentiment Composite ("DAVIS265");
   (g) *etfs*: Ned Davis Research's ETF Speculation indicator;
   (h) *sm2*: sentiment/momentum combined signal, defined as the change in the NDR DAVIS265 index over a lag period, multiplied by the difference between the SPY log return over the same lag period and the lag period; (Lansing, LeRoy, and Ma 2022);

3. **Variance-Related**:
   (i) *vix*: CBOE VIX volatility index;
   (j) *spxc2d*: "change times acceleration" (kinetic power), log change in SPY over its k_1-day value, multiplied by the second order change in SPY (log change in SPY vs its k_2 day value, minus the log change in SPY between its k_3 and k_2 days values);
   (k) *vs3*: composite volatility signal with three constituents: 1/ VIX, 2/ a variance risk premium metric (VIX squared minus the 20-day EMA of realized variance), and 3/ the change in VIX over its EMA times the change in SPY over its EMA;

4. **Seasonality and events**:
   (l) *com_tom*: Turn-of-month flag. Defining the last day of the month as T-1 and first day of next month as T+1, we go long at the end of T-2 and get flat at the end of T+3 and we go short at the end of T-8 and get flat at the end of T-4.
   (m) *fomc*: Indicator variable for FOMC meeting dates;
   (n) *sim*: "Sell In May" indicator variable (a.k.a. Halloween Indicator), which is 1 from mid-May to mid-October, and 0 otherwise (Andrade, Chhaochharia, and Fuerst 2013)

Transformations. As is typical with machine learning approaches, we transform our features in order to capture potential nonlinear relationships and reduce noise. These transformations must be performed in a walk-forward approach to avoid any potential leakage to the out-of-sample period.

We choose an expanding-window approach, where the minimum window for the in-sample period covers daily data over ten years, from 1990 to 1999. We find the optimal transformation for each feature independently. We create an expanded dataset from the transformations and find the optimized parameters for each transformation by minimizing the negative absolute correlation of the feature and the target (over the full in-sample data). We re-optimize every twelve months with the added data, and will use that new optimal transformation to predict the following year's excess return.

We consider the following transformations:

- *vs3*: Optimizing the weights of the three different components of the signal;
- *sm2*: Optimizing the length of the lag period;
- *BAA*: Difference between the current value and its lagged value, optimizing the lag period;
- *delinquency*: Indicator of whether the value is greater than the lagged value, optimizing the lag period;
- *spxc2d*: Optimizing the k_1, k_2, k_3 lag parameters;
- *eyby*, *termspread*, and *vix*: Discretized (-1, 0, 1) trading signal based on optimized thresholds maximizing the in-sample correlation with the target.

3.2 Methodology

Predictive regressions. We use an expanding sample approach, using 1990-1999 as the minimum in-sample period and refitting at the end of each month after that, with the added data. We fit an elastic net model (which combines the L1 penalty of LASSO and the L2 penalty of ridge regressions), with penalty coefficients found using 5-fold cross-validation[3]. The target is the daily return on SPY in excess of the Fed funds rate, while the predictors are z-scored candidate transformations of the variables described in the previous subsection. We use a 5-fold cross-validation for the elastic net

[3]Using a nested temporal CV does not seem critical here, given this is only performed in sample.

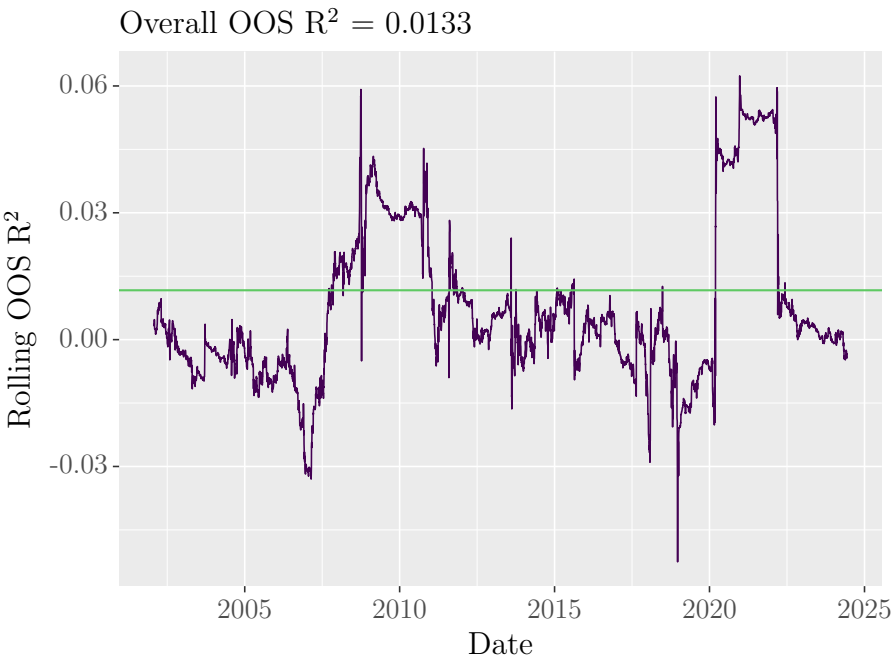
parameters over each in-sample period. Once again, we rely on an expanding-window approach for our in-sample dataset. While the transformations are re-optimized every year, the elastic net is refitted every month and used to predict the next month out-of-sample.

Model evaluation. We evaluate model performance in different ways. In the cross-validation, our main metric is the prediction RMSE of each test sample. For the out-of-sample (OOS) forecast, we compute an OOS R^2 using the average excess return over the in-sample period as our naive benchmark.

However, for practical purposes, we need to translate the daily forecast into a portfolio allocation. To do so, we use a simple heuristic where the percentage allocated to equity is $100\% + k \cdot \tilde{r}$, where k is our scaling factor. Given our goal to scale the signal between 0% and 150% equity, we found that $k = 500$ was a reasonable choice, although the results are qualitatively robust to other choices.

Based on this allocation, in the next section we will evaluate the strategy according to standard portfolio metrics: compound annual growth rate (CAGR), total return, annualized volatility, Sharpe and Sortino ratios, return-to-risk ratio, maximum drawdown, time under water, CAPM alpha and beta, hit rate, gain-loss ratio, and Treynor-Mazuy gamma.

Figure 1: Rolling out-of-sample R^2



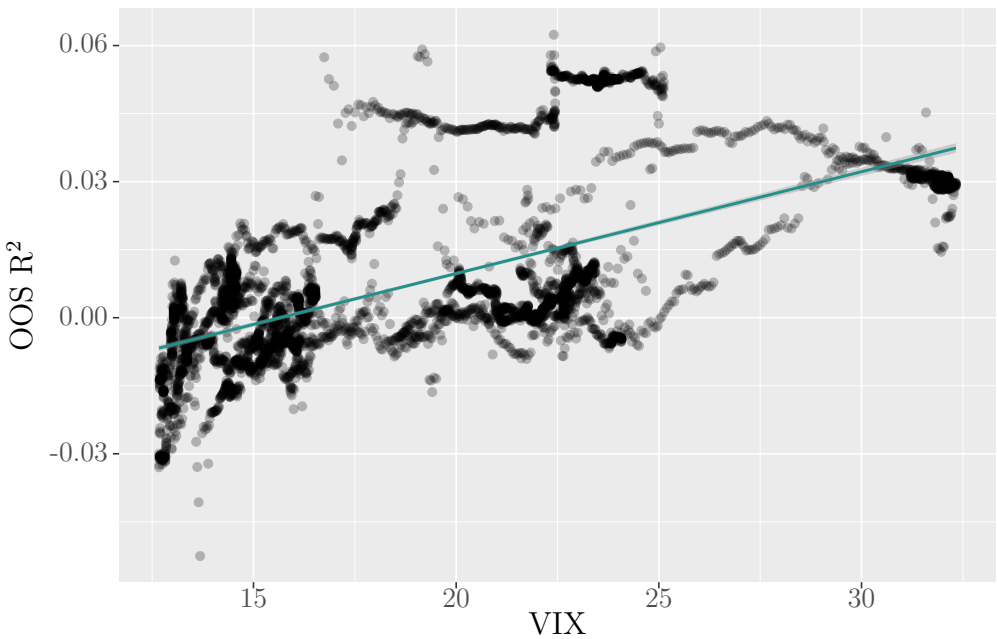
4 Results

4.1 Out-of-sample model fit

In order to assess the fit of the model, we look at the 2-year rolling out-of-sample (OOS) R^2 in Figure 1. The overall OOS R^2 of 1.33% from 2000 to 2024 translates into a roughly 10% correlation with the predicted value, which is fairly large for a one-day market prediction. However, looking at *when* the model performs better or worse than a long-term average is critical to assessing its usefulness.

The OOS R^2 is showing positive values throughout much of the sample period. In particular, the model achieves elevated R^2 levels (above 3%) during the 2008 financial crisis and again after 2020, indicating particularly strong predictive performance around and right after market crashes. In much of

Figure 2: Rolling out-of-sample R^2 vs VIX



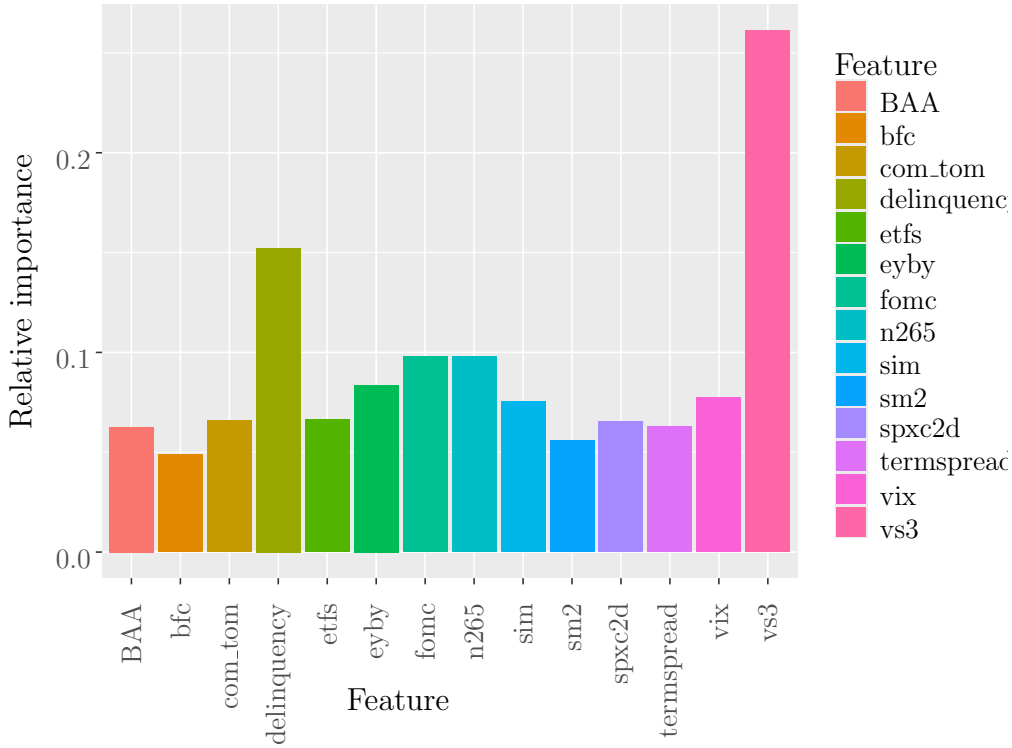
the other periods, the model fit as measured by the rolling R^2 remains near (but often above) zero. Because R^2 is not meaningful for measuring trading performance, we will look at how this affects the strategy across periods in the next subsection.

To better understand where the predictability comes from, we plot in Figure 2 the relationship between OOS R^2 and the level of VIX. Although there is a fair amount of noise, there is a general trend in which the model does worse in low volatility regimes and better in medium and high volatility periods. Most R^2 observations below -1% fall in the lowest VIX zone. It appears that the highest R^2 levels coincide with moderate volatility (VIX in the range of 18-25), though the converse is not true: some periods in that volatility range have near zero R^2.

Figure 3 shows the relative importance of the optimized features of the

Figure 3: Relative feature importance

This figure shows the relative feature importance for the model over the entire 2000-2024 period.

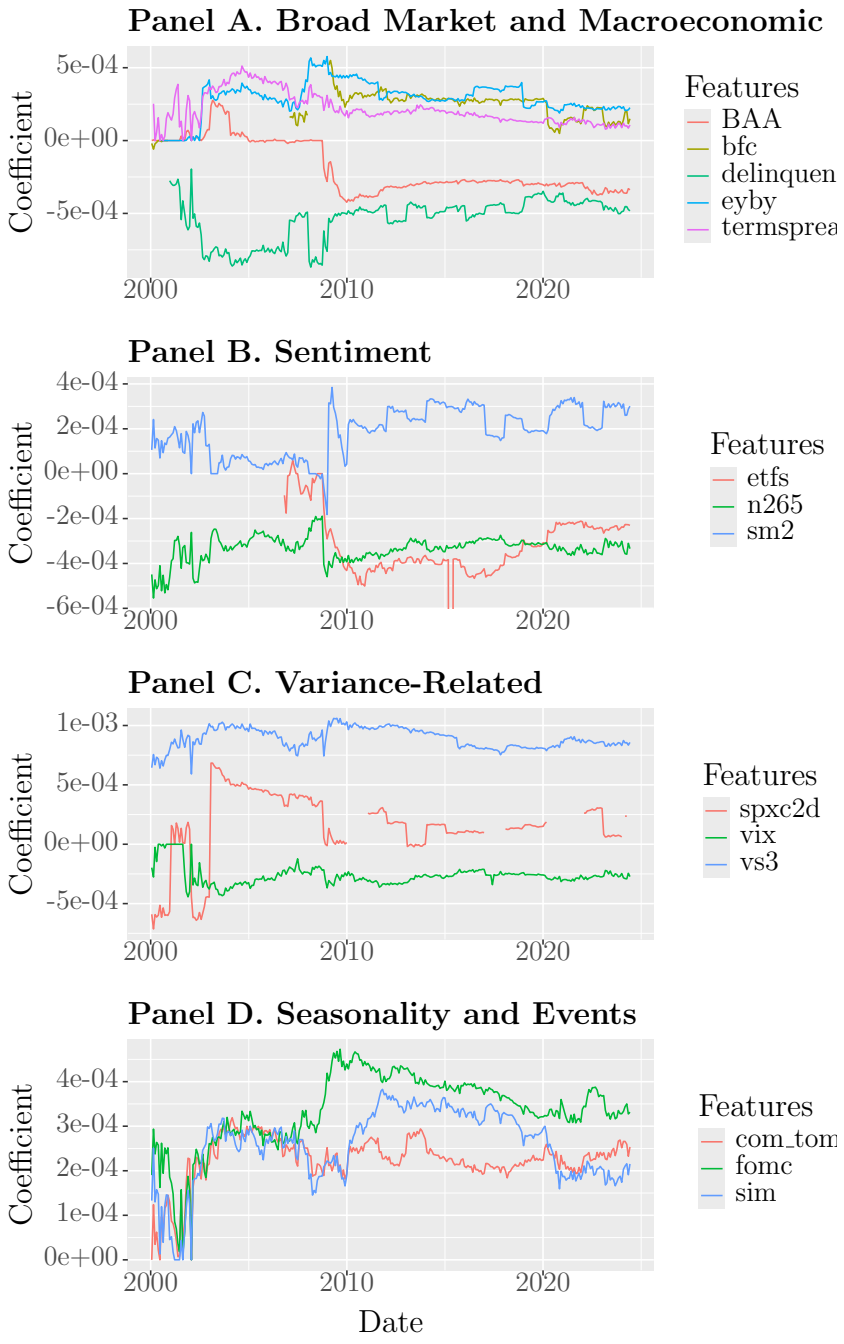


model throughout the sample, defined as the average absolute value of the coefficients of the model (based on z-scored variables). Most of the features scored similarly, with a couple of notable exceptions. First, the *vs3* volatility feature has almost three times the average importance and is, in fact, one of the most consistent features in our model. Second, *delinquency* comes out at twice the average score, probably because it effectively picks up the changes in the economy that led to the 2008 financial crisis.

Figure 4 presents the time-varying coefficients for model features grouped by category, covering the 2000-2024 sample, based on the monthly refits of

Figure 4: Model coefficients over time

This figure shows each feature's coefficient over the 2000-2024 sample.

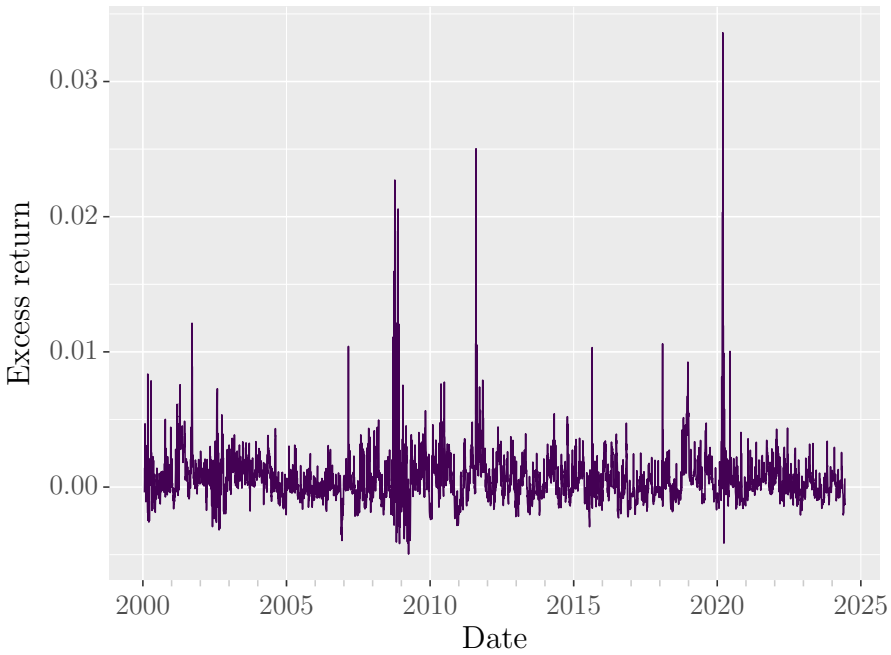


the elastic net model. In Panel A, macroeconomic indicators such as *BAA*, *bfc*, and *termspread* exhibit moderate fluctuations, with *BAA* and *termspread* showing more pronounced changes around the financial crisis and post-crisis periods. Panel B shows sentiment indicators, where the coefficients of *sm2* and *n265* are relatively stable, especially after 2010, while *etfs* (which only starts in 2008) has more volatility, particularly initially. Panel C highlights the variance-related indicators, with *vs3* maintaining a consistently high and positive influence, while the optimized *vix* transformation has a negative sign, and *spxc2d* comes in and out intermittently due to the L1 penalty. In Panel D, seasonality and events indicators reveal that *com_tom* and *sim* coefficients exhibit similar patterns early in the sample, but shift around the financial crisis, with *fomc* displaying a stronger and steadier trend post-2010. These dynamics suggest varying predictive importance across market conditions and time periods, reflecting how different economic signals influence the model's performance over time.

Figure 5 shows the raw predicted excess return signal generated by the model over the period from 2000 to 2024. The predicted excess return exhibits a few noticeable spikes, particularly during periods of high market volatility, such as around the 2008 financial crisis and the COVID-19 pandemic in 2020. These peaks suggest the model is picking up higher market opportunities in the crisis periods. Interestingly, most of the spikes are positive, and the model rarely, if ever, predicts large negative drops. This suggests the model profits by buying dips rather than avoiding them.

In Figure 6, we observe the predicted signal scaled to determine an equity weight allocation for SPY, which oscillates in a range of 0 to 1.5. The frequent fluctuations in weight reflect the model's responsiveness to daily market signals, adjusting exposure dynamically to optimize risk and return in response to changing conditions. Naturally, this sometimes requires large rebalancing from day to day. We penalize them using transaction costs of 0.6 cents per share, and we also assume a 0.25% credit spread to the benchmark rate.

Figure 5: Predicted excess return

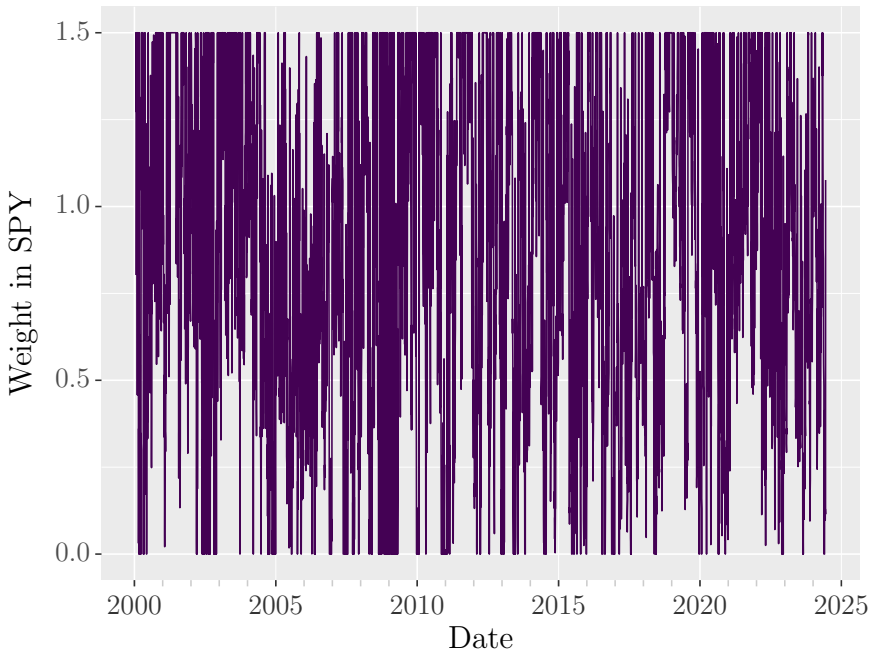


4.2 Strategy performance

The full-sample results presented in Table 1 underscore the superior performance of the strategy relative to the benchmark across several key performance metrics, highlighting its ability to generate higher returns and risk-adjusted efficiency. The compound annual growth rate (CAGR) of the strategy (14.0%) substantially exceeds that of the benchmark (7.5%), resulting in a total return more than four times higher (2334% vs. 484%). This considerable disparity suggests that the strategy effectively capitalizes on growth opportunities, outperforming the benchmark over the observed period.

Volatility, while somewhat elevated for the strategy (21.8% vs. 19.4%), does not appear to undermine its effectiveness in managing extreme losses. Risk-adjusted measures, including the Sharpe ratio (0.556 vs. 0.294) and Sortino ratio (1.074 vs. 0.663), indicate that the strategy delivers returns

Figure 6: Scaled weight in SPY



with a higher risk efficiency. The Sharpe ratio, measuring excess return per unit of volatility, reflects that the strategy yields 88.8% more return per unit of risk than the benchmark. Similarly, the Sortino ratio, which accounts only for downside volatility, confirms the strategy's superior downside risk management. The Return/Risk metric (0.643 vs. 0.386) further corroborates this finding, showing that the strategy achieves a higher proportionate return relative to total risk exposure.

The strategy's maximum drawdown (38%) is significantly lower than that of the benchmark (55%), signaling enhanced resilience during market downturns. This resilience is further emphasized by the shorter time under water (1,188 days vs. 1,657 days), suggesting quicker recovery periods and less prolonged periods of drawdowns compared to the benchmark.

The strategy's positive CAPM alpha (6.406%) and beta close to unity

Table 1: Performance Statistics (Full Sample)

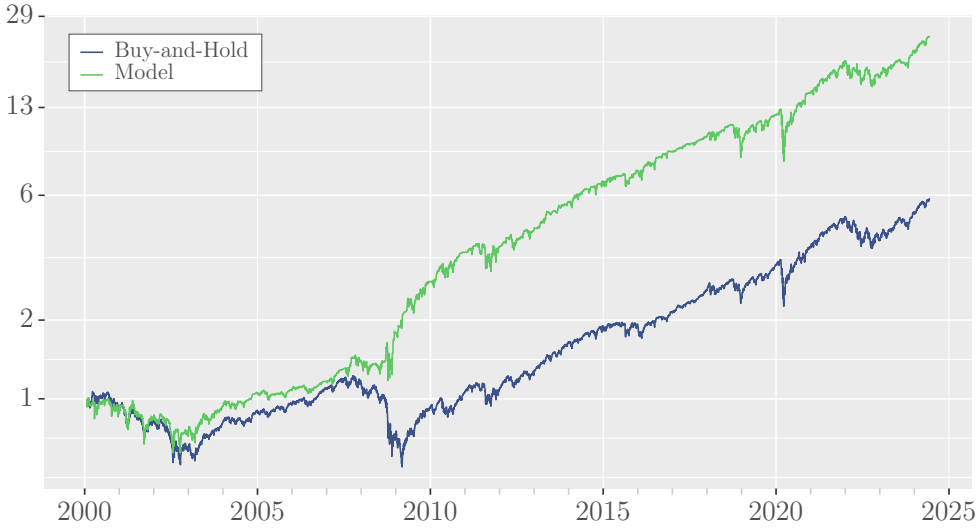
	Strategy	**Benchmark**
CAGR (%)	14.012	7.515
Total Return (%)	2334.733	483.654
Volatility (% Ann.)	21.801	19.454
Sharpe	0.556	0.294
Sortino	1.074	0.663
Return/Risk	0.643	0.386
Max. Drawdown (%)	38.461	55.201
Time Under Water (days)	1188.000	1657.000
CAPM α (% Ann.)	6.406	
CAPM β	0.990	
Hit Rate (%)	52.681	
Gain Loss Ratio	1.180	
Treynor-Mazuy Gamma	3.078	

(0.99) suggest that, while closely tracking the benchmark's market exposure, the strategy consistently generates excess returns beyond what is expected based on its systematic risk exposure. The significant Treynor-Mazuy gamma (3.078) further implies that the strategy captures some nonlinear returns, potentially indicating effective market timing or dynamic risk management capabilities.

Additional performance metrics, including the hit rate (52.7%), demonstrate that the strategy consistently generates positive returns and is more frequently correct in directional moves than incorrect. Finally, the gain-loss ratio of 1.180 indicates that, on average, profitable trades outweigh losses, both in magnitude and frequency.

In summary, the strategy achieves notable outperformance relative to the benchmark, evidenced by higher returns, superior risk-adjusted performance, enhanced downside protection, and a positive CAPM alpha. This suggests the strategy may effectively exploit market inefficiencies, yielding superior outcomes for investors with acceptable levels of risk exposure. Figure 7 illustrates the difference in cumulative wealth of the strategy compared to the benchmark.

Figure 7: Wealth accumulation (2000-2024)



5 Robustness checks

5.1 Subsample Analysis: 2010-2024

A subsample analysis of the 2010-2024 period is presented in Table 2. Compared to the full 2000-2024 sample, the strategy still displays robust performance in excess of the benchmark, though with slightly varying risk-return dynamics that reflect the different market environments characterizing each sample. Notably, the strategy's CAGR improved from 14.0% in the full sample to 16.1% in the 2010-2024 period, suggesting that it capitalized on favorable trends and growth opportunities in the post-crisis bull market, which was marked by quantitative easing and low-interest-rate policies that fueled asset price appreciation. It is also plausible that the longer in-sample period leads to better statistical robustness in the model. The return on the benchmark improved dramatically after the financial crisis, rising from 7.5% to 13.5%, but the strategy maintained a consistent outperformance.

Table 2: Performance Statistics (2010-2024)

	Strategy	**Benchmark**
CAGR (%)	16.157	13.549
Total Return (%)	766.458	524.574
Volatility (% Ann.)	19.891	17.165
Sharpe	0.752	0.721
Sortino	1.230	1.158
Return/Risk	0.812	0.789
Max. Drawdown (%)	36.796	33.700
Time Under Water (days)	384	481
CAPM α (% Ann.)	2.009	
CAPM β	1.058	
Hit Rate (%)	53.931	
Gain Loss Ratio	1.220	
Treynor-Mazuy Gamma	1.282	

Risk-adjusted performance metrics such as the Sharpe and Sortino ratios further corroborate the robustness of the strategy. The strategy's Sharpe

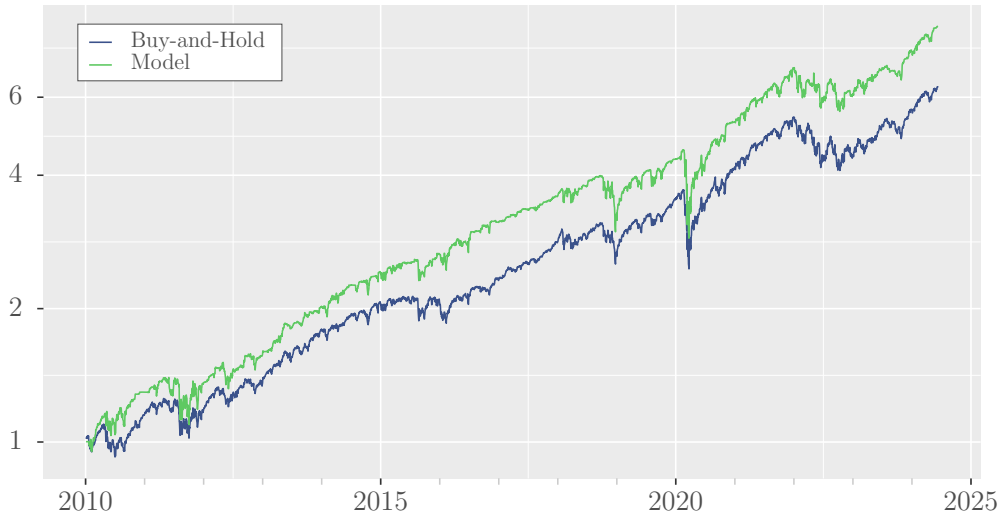
ratio rose from 0.556 in the full sample to 0.752 in the 2010-2024 period, while the benchmark's improvement was more moderate (from 0.294 to 0.721), illustrating the strategy's enhanced efficiency in converting risk into returns during the latter period. Similarly, the Sortino ratio increased from 1.074 to 1.230, suggesting that the strategy was particularly effective at managing downside volatility in the less turbulent post-crisis years.

Despite a reduction in maximum drawdown from 38.5% in the full sample to 36.8% in the recent period, the strategy preserved its resilience relative to the benchmark, which exhibited a more substantial reduction in maximum drawdown (from 55.2% to 33.7%). This shift underscores a moderating effect in systemic risk post-2008, as both the strategy and benchmark experienced less pronounced drawdowns in the subsequent period.

The CAPM alpha, an indicator of risk-adjusted excess return, decreased from 6.4% in the full sample to 2.0% in the recent period, reflecting a diminished but still positive advantage over the benchmark as market returns grew. This decline likely reflects both the rise in market beta (from 0.990 to 1.058) and the narrowing of the opportunity for capturing alpha in a more stable economic environment. Despite this, the strategy's Treynor-Mazuy gamma remained positive, indicating potential nonlinearities are playing a role, and further reinforcing the robustness of the strategy's performance and adaptability to evolving market conditions.

Overall, the 2010-2024 results, while demonstrating slightly lower alpha and less extreme drawdowns, indicate a consistent outperformance of the strategy across multiple market environments. These results underscore the strategy's robustness, maintaining advantageous risk-return trade-offs and resilience under distinct economic phases, and provide strong evidence of its sustained effectiveness relative to the benchmark over time. Figure 8 demonstrates the cumulative effect of the strategy's outperformance in the 2010-2024 subperiod.

Figure 8: Wealth accumulation (2010-2024)



5.2 Subsample Analysis: 2019-2024

Now focusing on an even shorter sample in the 2019-2024 period, the strategy summarized in Table 3 demonstrates resilient and adaptable performance amid varying market conditions, including the heightened uncertainty of the COVID-19 era and subsequent recovery. Over the 2019-2024 period, the strategy's CAGR climbed to 19.6%, representing a notable increase relative to both the 2010-2024 period (16.2%) and the full sample (14.0%). This enhanced growth rate, paired with a benchmark CAGR of 17.1%, indicates that both the strategy and benchmark thrived in a largely bullish environment, though the strategy continued to maintain its competitive edge. The wealth accumulation of the strategy and its comparison to the benchmark is illustrated in Figure 9.

Volatility for the strategy also rose during this period (23.2%), reflecting

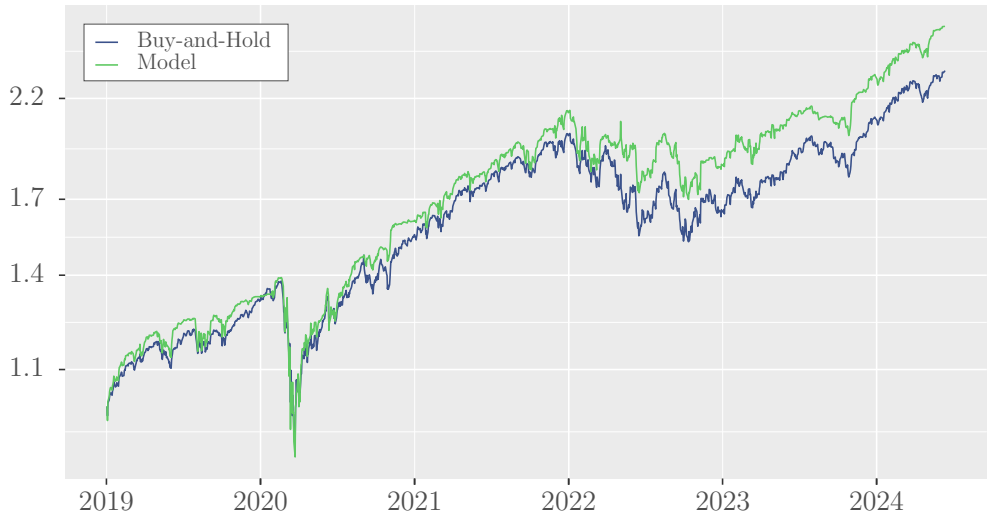
Table 3: Performance Statistics (2019-2024)

	Strategy	**Benchmark**
CAGR (%)	19.587	17.122
Total Return (%)	164.248	135.981
Volatility (% Ann.)	23.167	20.356
Sharpe	0.738	0.721
Sortino	1.281	1.233
Return/Risk	0.845	0.841
Max. Drawdown (%)	36.796	33.700
Time Under Water (days)	384.000	481.000
CAPM α (% Ann.)	2.042	
CAPM β	1.041	
Hit Rate (%)	54.480	
Gain Loss Ratio	1.231	
Treynor-Mazuy Gamma	1.240	

a slight uptick from prior intervals, potentially driven by pandemic-induced market swings. Despite the increased volatility, the Sharpe ratio of 0.738 demonstrates that the strategy managed risk efficiently, maintaining a similar level of risk-adjusted performance compared to the 2010-2024 period (0.752) and considerably exceeding the full-sample Sharpe (0.556). Additionally, the strategy's Sortino ratio increased to 1.281, indicating strong downside protection even in this higher-volatility period, with the benchmark showing a comparable Sortino of 1.233. The maximum drawdown remained steady at 36. 8% in the two more recent periods. The benchmark similarly maintained consistent risk levels, with its drawdown at 33.7% in 2019-2024.

Lastly, the CAPM alpha remained around 2.0% in the most recent period. The strategy continued to deliver positive alpha, confirming its ability to achieve risk-adjusted returns beyond market expectations. In particular, the hit ratio of the strategy reached 54.5% in 2019-2024, an improvement over previous periods, suggesting a better ability to capture favorable market opportunities during recent volatility. The Treynor-Mazuy Gamma of 1.240 remained largely in line with that of the 2010-2024 sample.

Figure 9: Wealth accumulation (2019-2024)



Overall, the 2019-2024 results illustrate the strategy's resilience, maintaining consistent risk-adjusted outperformance and confirming its robustness across varied economic conditions, from long-term growth to short-term volatility spikes.

6 Conclusion

In this paper, we introduce a framework leveraging micro alphas—weak but collectively informative signals—to generate out-of-sample predictability for the equity risk premium. Our approach draws from machine learning techniques, notably elastic net regularization, to identify and aggregate these signals, managing multicollinearity and nonlinear relationships that are otherwise challenging to address in traditional financial models. The strategy demonstrated robust performance metrics, with a full-sample Sharpe ratio of 0.556, Sortino ratio of 1.074, and CAPM alpha of 6.4% indicating significant risk-adjusted returns relative to a traditional equity benchmark. During periods of elevated market stress, the predictive power of our model intensifies, with the strategy achieving particularly high out-of-sample R^2 values during the 2008 financial crisis and COVID-19 pandemic, reinforcing our hypothesis that market turbulence can enhance the visibility of predictive inefficiencies.

Market timing has traditionally been met with skepticism, often dismissed as speculative or impractical. However, we anticipate a shift in this narrative, where failing to incorporate market timing into an investment strategy could increasingly be seen as a missed opportunity or even a lapse in responsible portfolio management. A natural continuation of this research involves the ongoing search for and validation of additional micro alphas, as market evolution and structural changes may give rise to new predictive signals while potentially diminishing the effectiveness of others. The framework could also be extended to multi-asset portfolios, where the potential for micro alphas could vary significantly with different return profiles and liquidity conditions. Another promising avenue involves integrating alternative data sources, such as sentiment from social media or macroeconomic forecasts analysis from large language models, to capture additional forms of informational advantage. By exploring these areas, future researchers as well as practitioners may refine this method for tactical asset allocation, and find adaptive performance as the market evolves.

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